
Build A CubeSat: A Collaborative, Hands-On Program for High School Students at MIT

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SmallSat Education Conference 2023

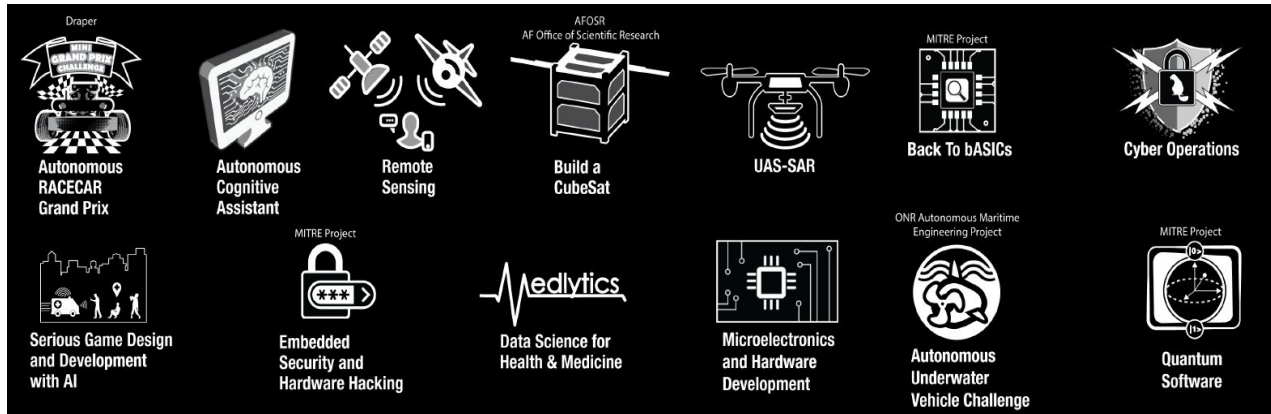




MIT Beaver Works Summer Institute



- **BWSI was started in 2016 to fill a need :**
Project-based STEM curriculum with focus on AI and CS skills
- **BWSI Core :**
 - No cost to students, any talented and passionate students able to attend
 - Project-based with teamwork
 - Scientists and engineers actively working in research as instructors
 - Teams are motivated by competitions and capstones projects
 - Participation is based on independent (online) program prior to 4 week program

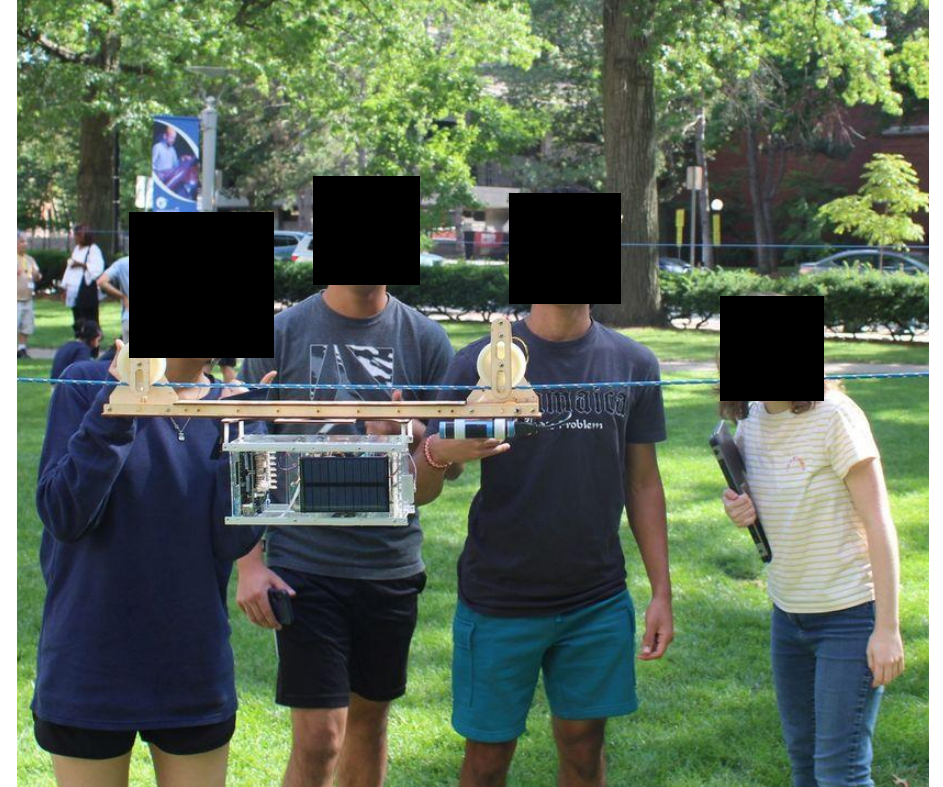




Build A CubeSat Course



- The Build A CubeSat course invites students to design and build their own prototype CubeSat
- 4 week full time summer program focuses on hands-on project based learning mixed with lectures on satellite systems from subject matter experts
- Partnership with Woods Hole Oceanographic Institution (WHOI)
 - Support hardware kit develop and provide yearly science mission
- CubeSats are a tool to motivate learning:
 - Fundamental engineering skills (intro to EE, ME, CompSci, and Aerospace)
 - Teaming, peer leadership, and project management
 - Communication (documentation, presentations, and daily coordination)

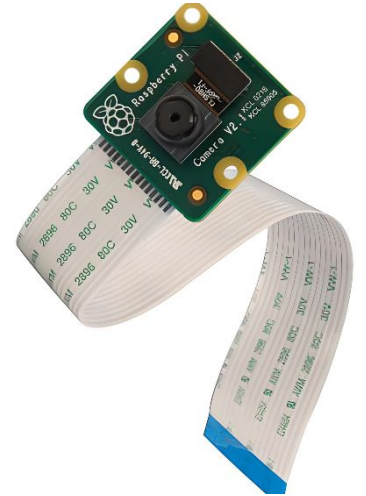
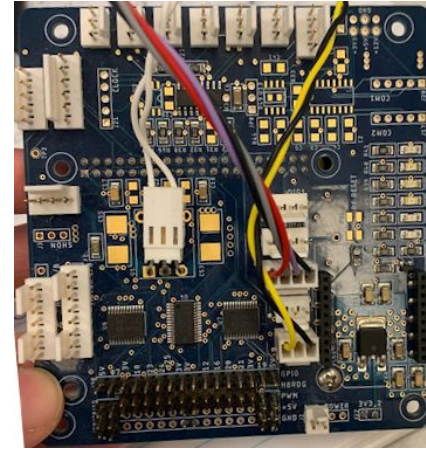




The Hardware



- Each student was given their own Raspberry Pi to use for the length of the course.
- On team levels, they were also provided with XBee radios, IMUs, Raspberry Pi cameras, solar cells, GPSs, batteries, and a custom interface board called the Chicken Bot Pi.





How do you teach high schoolers to Build a CubeSat in one month?



- **Focus on emotional and social problems just as much as technical ones**
- **Set up technical challenges in a way that the students learn progressively in order to build comfort and confidence**
- **Form teams that both support and challenge students**

The program can only be a success if students are both learning AND having a good time.



Course Assessment



How much did you enjoy the following components of the course

15 Strongly Disliked Disliked Neutral Enjoyed Strongly Enjoyed

How much did you learn from the following components of the course

Nothing A Little A Lot This was life changing

- **Each aspect of the course is individually assessed on both the student enjoyment as well as the student learning**
- **A post-course survey is our primary source of information, in which 14/19 of the students this summer participated**
- **Students scored every part of the class and provided short answer feedback**

We will also share direct quotes from students.



Course Outline



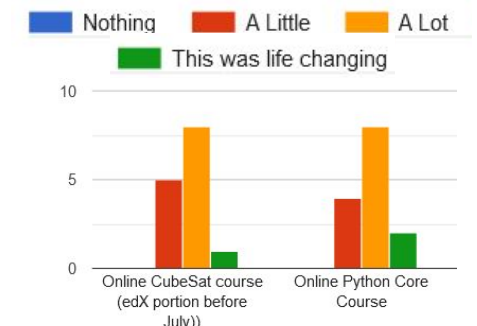
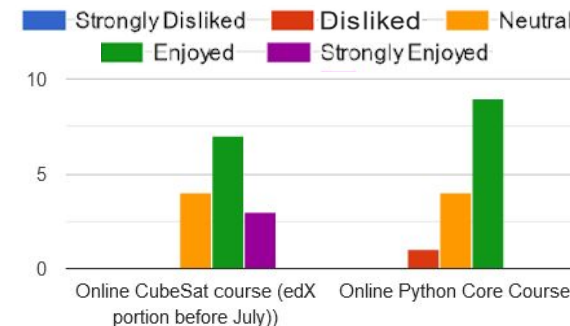
Application, including online course					
	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	Intro	Systems Engineering	FlatSat Scavenger Hunt	Mission Requirements	Science Mission
	FlatSat Challenge			Meet the Hardware	
Week 2	Communications	Structures + Imaging	ADCS	Power	Software and CONOPs
	Subsystem Labs				
Week 3	Propulsion + Ground Stations	MIT Lab Tours	Thermal + Risk	Guest Lecture	Design Review
	Final Project				
Week 4	TROPICS	Test Day	Test Day	Flight Day	Final Event Prep
	Final Project				



Pre Course Material



- **Self-paced Edly courses in satellite engineering concepts, Systems Tool Kit, Python, and GitHub**
- **Provide applicants a Piazza to discuss course material with other students**
- **Not only is the prerequisite information useful for the course, but it helps students learn to solve problems without a teacher to give a solution.**
- **Well-written piazza answers and discussions are a great way to help choose students to be accepted!**





Lectures and Seminars



- Every morning began with 1-2 lectures
- Each lecture had a different speaker, who was either an engineer from MIT Lincoln Lab, an MIT professor, or an MIT graduate student
- The lectures were full of college-level material that covered both technical and systems engineering topics, many of which the students have never seen before.
- Seminars from different researchers and industry professionals were held daily across all of BWSI.



“I really liked that every day I was learning something new everyday with the seminar.”



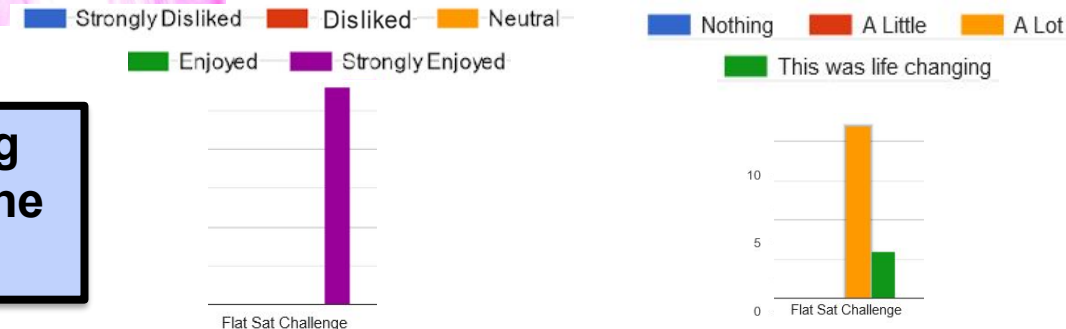
FlatSat Challenge



- The first team project!
- Program the Raspberry Pi to take a picture when it is shaken by using the IMU
- Compete in a scavenger hunt around campus
- Teaches students basic code integration and the basic functions of the Raspberry Pi, the camera, the IMU, and GitHub
- Gives students early victories, teamwork, and bonding



“First awesome thing we did, running around campus taking pictures was really fun, work with your first team, starts off the courses really well.”





FlatSat Challenge

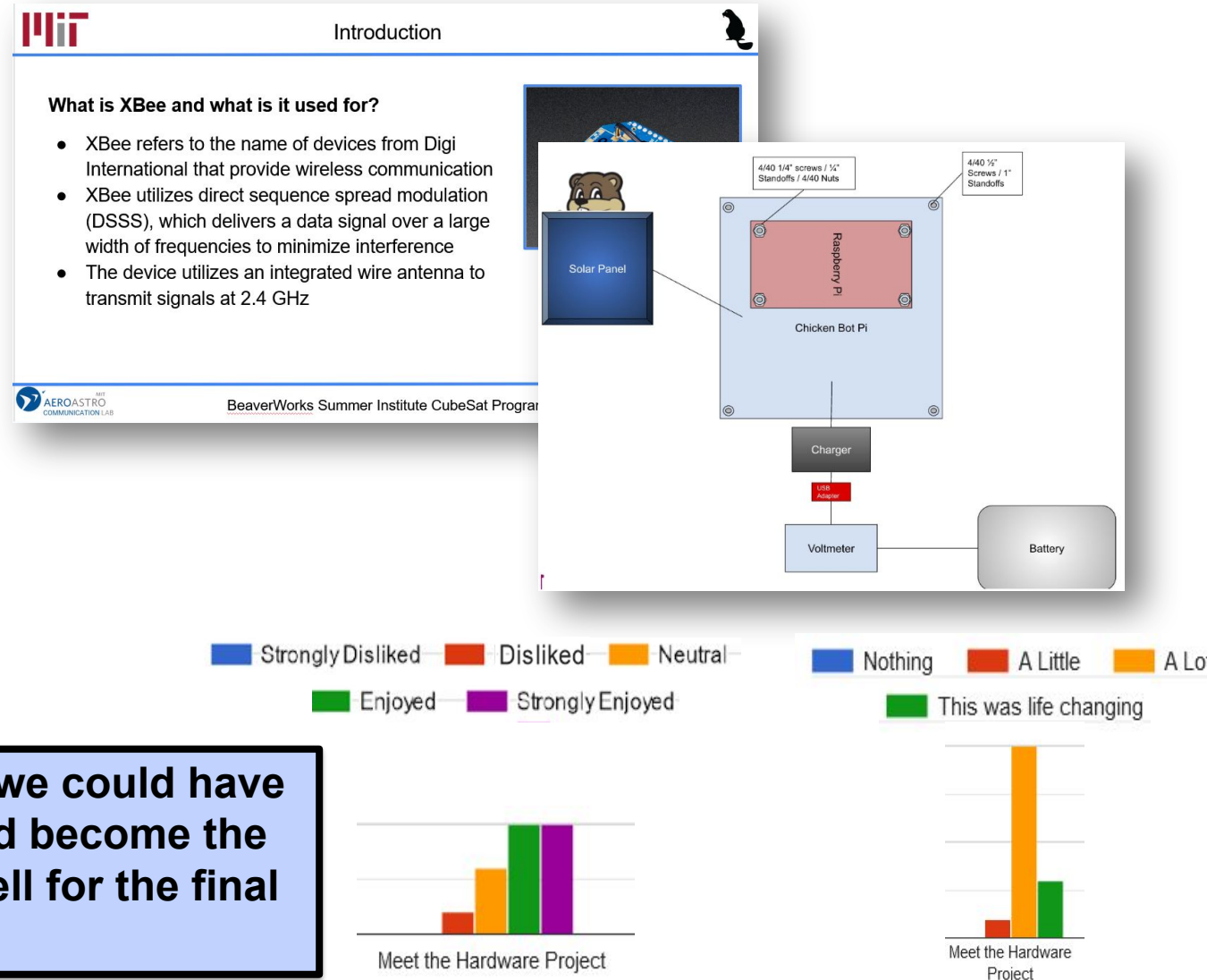




Meet the Hardware Project



- Put the students into groups to learn one of each of the CubeSat subsystems and give a presentation to explain the requirements for the subsystem and how to use the hardware.
- Teaches students about learning from datasheets and circuit diagrams, writing instructive code, writing requirements, slide design, and presenting.
- Very important to give each student a baseline assessment of their presentation skills!



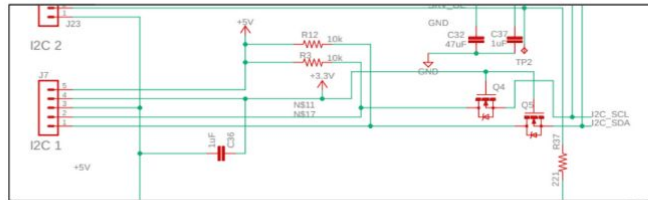
"It was super useful doing Meet the Hardware so we could have the chance to dig into a particular subsystem and become the "expert" on it. Meet the Hardware prepared me well for the final project."



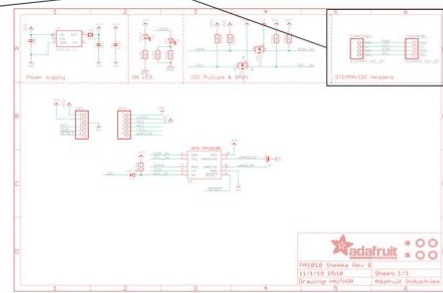
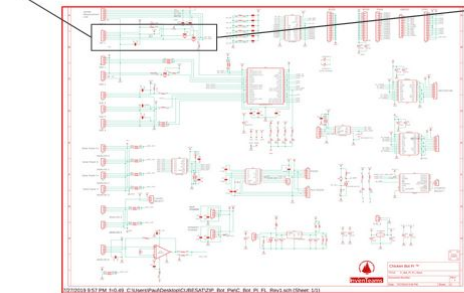
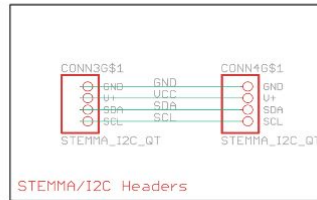
Meet the Hardware



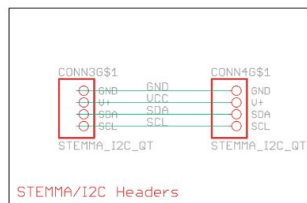
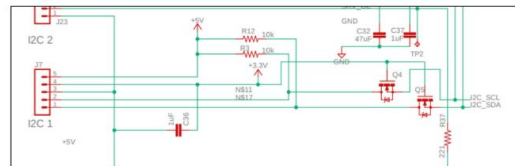
Chicken Bot Pi Schematic



GPS Schematic



Chicken Bot Pi integration



I2C connections (CBP pin to quick connect)

1. → SDA connection (quick connect 3)
2. → SCL connection (quick connect 4)
3. → Ground (quick connect 1)
4. → 3.3V power (quick connect 2)



Requirements



ID	Requirement Text	Rationale	Verification Type	Parent Req ID
MIS_REQ_01	The CBP shall have a reliable source of power.	The CBP needs power to work.	Test, Verification	
SYS_REQ_01	The power system shall have solar charging capabilities.	Harnessing solar energy for use in powering Raspberry Pi.	Test	MIS_REQ_1
SYS_REQ_02	The power system shall supply 2500 mAh of current and 10 Wh of power to store in the battery.	Receiving large amount of current will allow for easing of power budgetting; more time can be allocated towards scientific mission.	Test	MIS_REQ_1
SYS_REQ_03	The power system shall be able to use all of the generated power.	The system should use all of the energy it receives for efficiency	Test, Verification	MIS_REQ_1
SYS_REQ_04	The power system shall be able to monitor battery charge level and power draw information	It is helpful to check the charge level and power draw	Test	MIS_REQ_1
SYS_REQ_05	The power system shall have protective mechanisms to prevent overcharging, discharging, and short circuits	This would prevent anything harmful from happening	Inspect	MIS_REQ_1
SYS_REQ_06	The power system shall be able to generate power using a solar panel and store it in a battery	This will allow the system to be continuously operational from the battery storage	Inspect	MIS_REQ_1
COM_REQ_01	The battery shall have a high capacity.	More power can be stored and it can charge the CBP for long periods of time	Test, Verification	SYS_REQ_1
COM_REQ_02	The battery shall be capable of providing the output required by the CBP.	The CBP needs a specified amount of power and the battery should be able to provide it	Test	SYS_REQ_06
COM_REQ_03	The battery shall include protection circuitry to prevent voltage from going too high/low, and short circuits.	There needs to be safety requirements in place to prevent harm from the battery	Inspect	SYS_REQ_05



BeaverWorks Summer Institute CubeSat Program

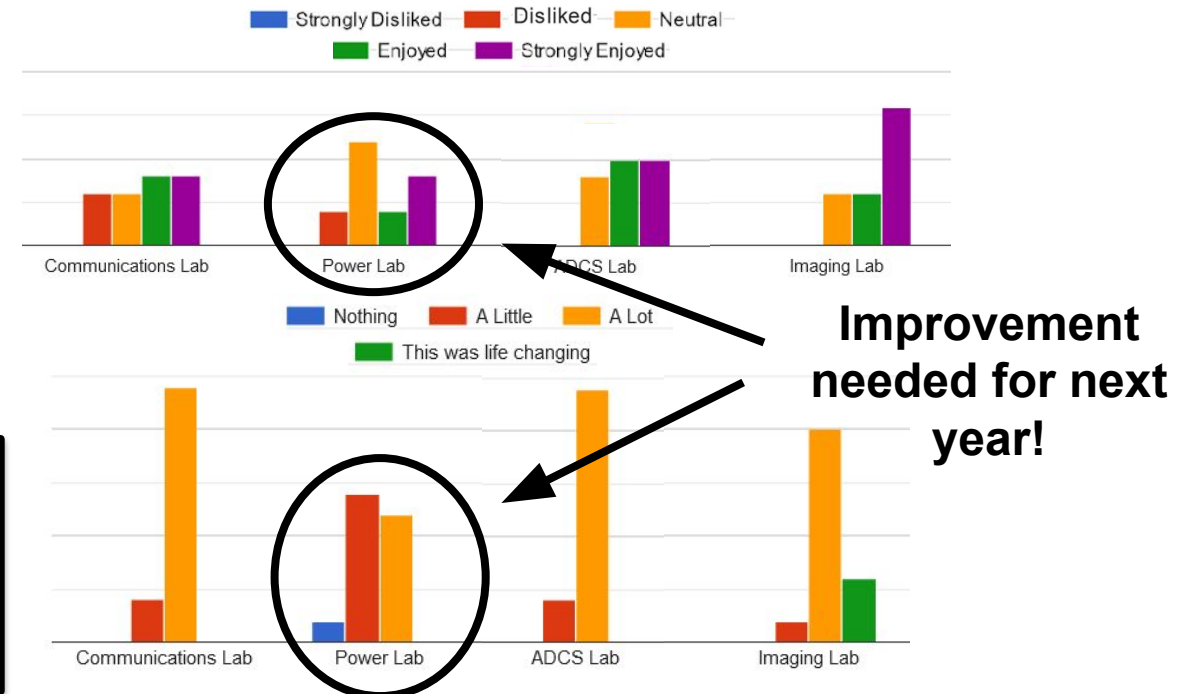


Subsystem Labs



- Each piece of hardware on the CubeSat had a corresponding lab, teaching the students skills they needed for the final project.
- Continually swapping teams forces students to all learn how to work together.

“Being forced to be in a team that I would not have normally chosen really was amazing because I was able to become close and work with people who I wouldn't have initially chosen to be on a team with.”

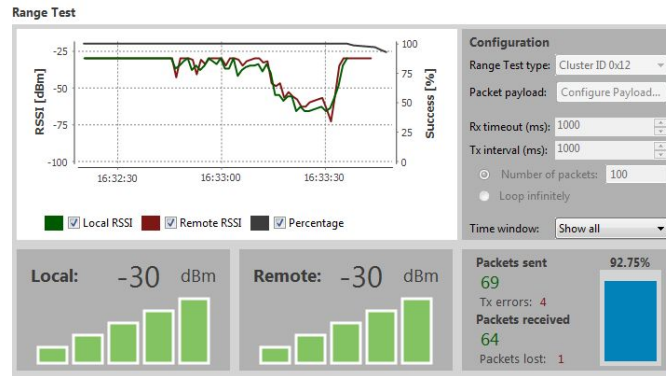




Subsystem Labs



Communications



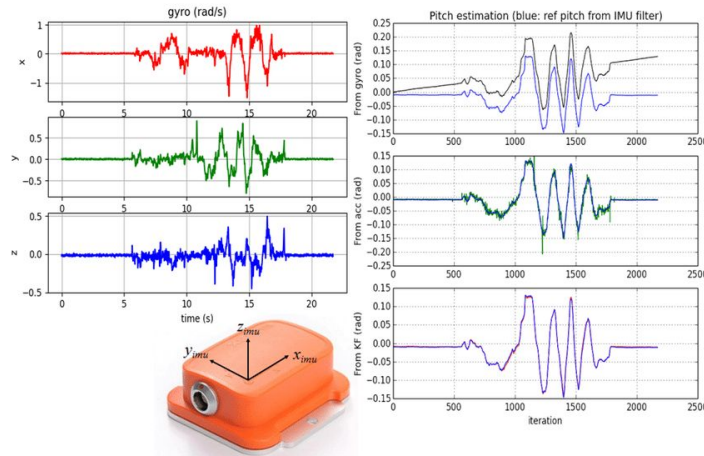
Send messages across distances and learn how packets and data loss work

Power



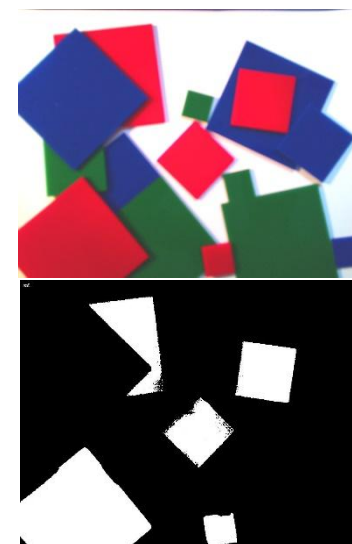
Make a power budget and measure the change in power generation with different angles of the sun on solar panels.

ADCS



Calculate and plot roll, pitch, and yaw with different sensors on IMU and compare them.

Imaging



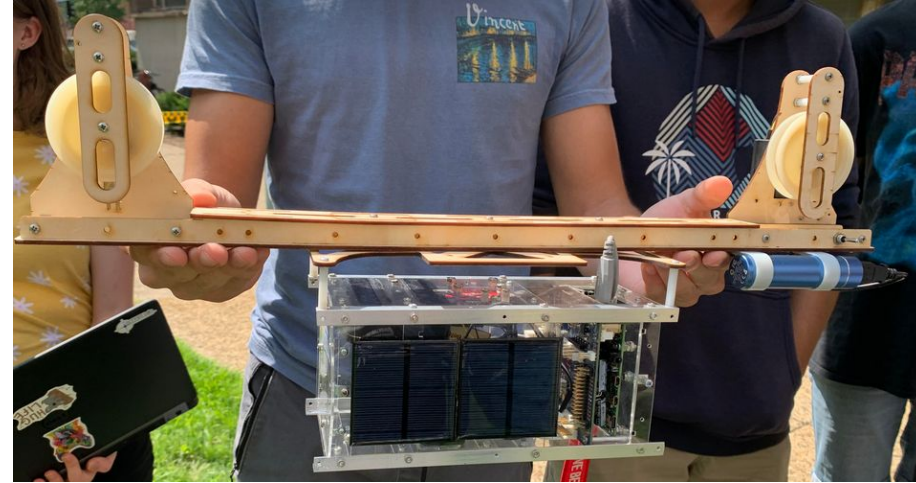
Experiment with image settings and write color segmentation code to count the amount and colors of and brightness of LEDs.



Final Challenge

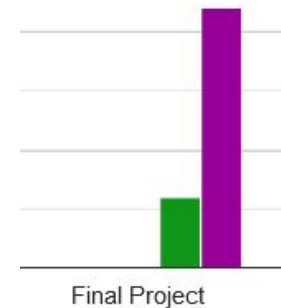


- **Create a CubeSat that will:**
 - Go down a zipline
 - Take pictures of LED cities
 - Send the images through a radio
 - Calculate position with the IMU
 - Calculate the color and brightness of the images
 - Maintain an accurate and positive power budget.
- **Students had creative freedom over the structure design, software architecture, CONOPs... basically everything!**
- **Students were scored on how well they performed.**

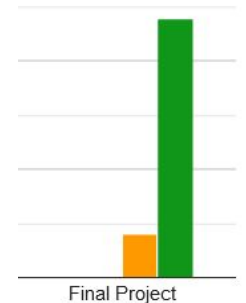


“The final project...showed us how important the systems part of engineering is in planning everything out. It also simulated how situations can really change and how quickly we have to adapt.”

Strongly Disliked Disliked Neutral
Enjoyed Strongly Enjoyed



Nothing A Little A Lot
This was life changing





Design Review



- **Students all gave presentations on their CubeSat design, building on their presentation skills.**
- **Forces the students to take the time to understand and explain everything instead of just pushing through with disorganized work.**
- **Allows the staff time to ask tough questions and provide serious feedback to the students.**

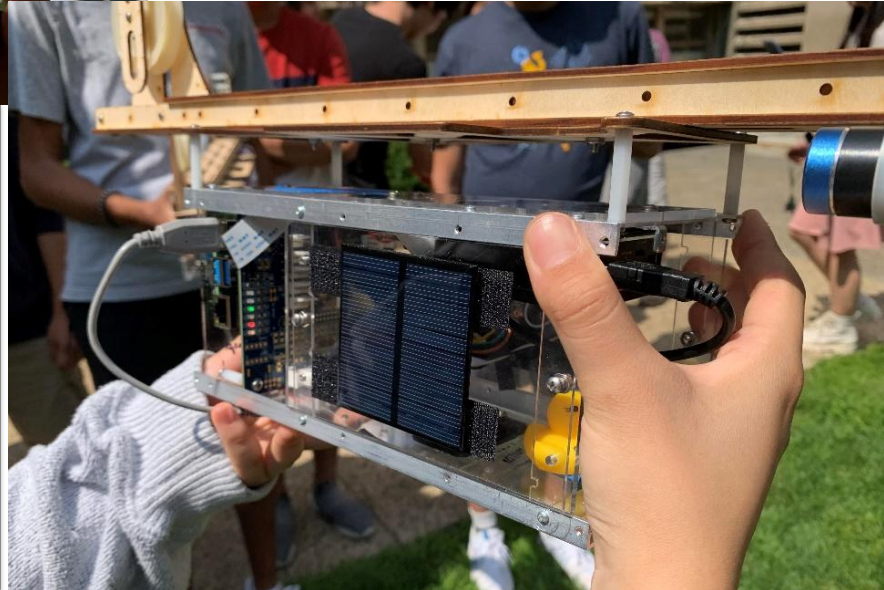




Flight Day



“The night before test day, almost everyone stayed back and was super supportive of all the other groups. Everyone made so much progress and no group would have done nearly as well without the other groups pushing them.”





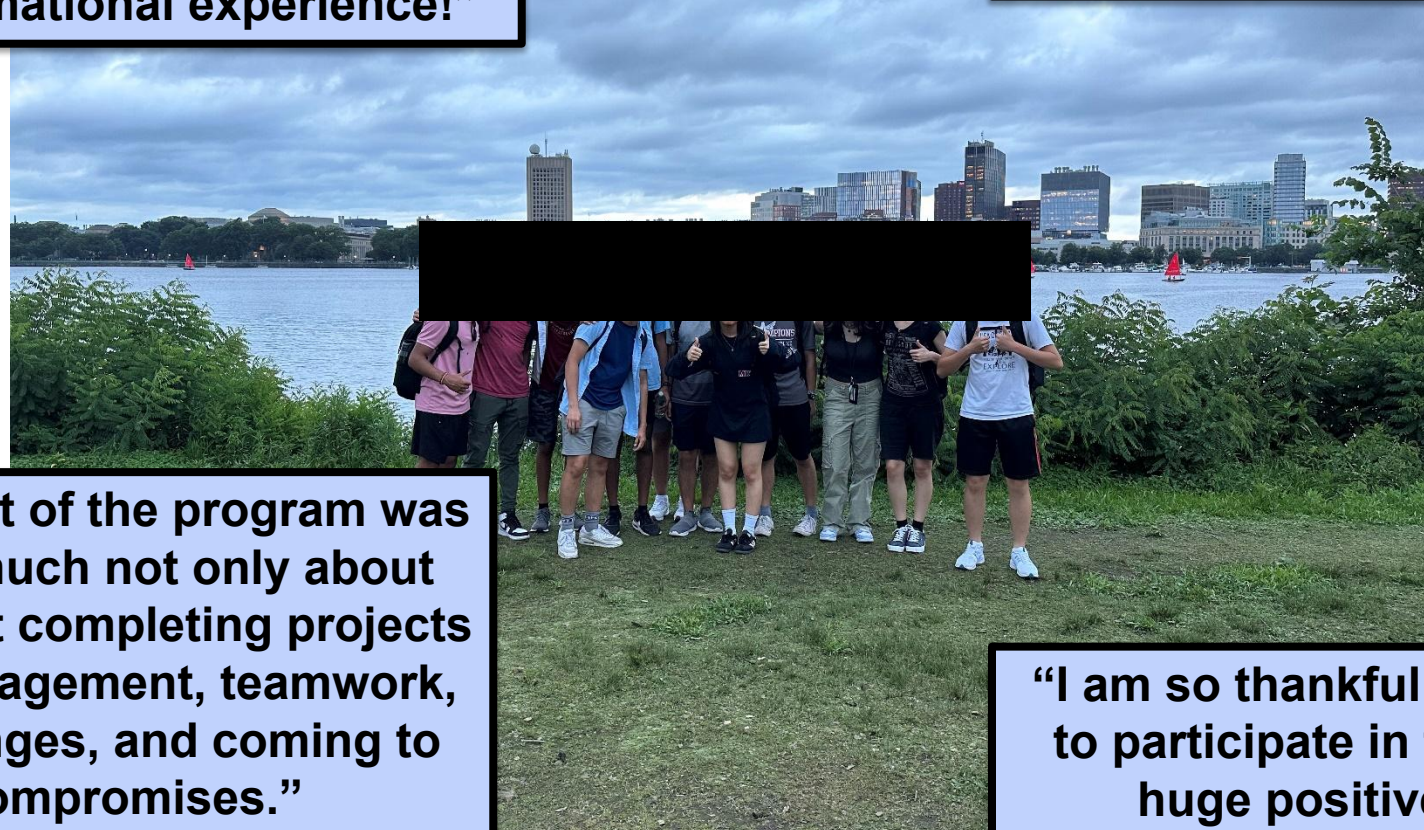
Student Retrospectives



"I will never forget this summer I spent here at MIT. I've met so many good people and learned so much."

"Thank you for a transformational experience!"

"I learned many engineering skills that will be very beneficial to me in college and career. And, it was such a fun and supportive environment!"



"The teamwork aspect of the program was great. I learned so much not only about engineering but about completing projects in general - time management, teamwork, working with challenges, and coming to agreements/compromises."

"I am so thankful I had the opportunity to participate in this course - it had a huge positive impact on me."



Thank you for listening!



Learn more about Beaver Works and how to apply to future BWSI programs:

<https://beaverworks.ll.mit.edu/CMS/bw/bwsi>

Apply to or mentor for the Build a CubeSat Challenge:

https://beaverworks.ll.mit.edu/CMS/bw/BWSI_BAC_Challenge

Contact bwsi-admin@mit.edu